

Order, cost and stability in diffusion sampling

A numerical-analysis audit of DPM-Solver across three tiers of testbed

CSE-402 Numerical Analysis, Simulation & Modeling — Section A

Abstract

Diffusion image sampling is an initial-value problem: one integrates the probability-flow ODE from noise to data, paying one neural-network evaluation (NFE) per stage. Lu et al. (*DPM-Solver*, NeurIPS 2022) accelerate this by solving the linear half of the ODE exactly, and validate the result with FID — an image-quality score. We grade the same solvers by the standards of numerical analysis — measured convergence order, stability at the stiff boundary $t \rightarrow 0$, and error per NFE — on three testbeds of decreasing reference exactness (an anisotropic Gaussian with a closed-form trajectory, a point mixture referenced against DOP853, and the real CIFAR-10 DDPM network referenced against a fine grid of itself), with three solver arms that separate DPM-Solver’s two ideas.

The order theorem holds where its hypotheses do: on the exact-score testbeds, in the asymptotic regime, every solver reaches its textbook order to within 0.094. But at the budgets the method is actually used at (10–120 NFE, down to $t_{\text{end}} = 10^{-3}$) no solver is in that regime — *even with an exact score*, measured order misses theory by up to 1.9, most of the 2.8 shortfall seen on the real network. Explicit steppers in t do destabilise near the data, but because of *curvature*, not stiffness: never on the linear testbed at any condition number, while on the curved mixture RK4 in t has a real stability limit that the λ -reparameterisation removes. The paper’s unexplained Table 6 anomaly — order 3 worse than order 1 at ~ 10 NFE — reproduces on the paper’s own checkpoint: at matched NFE, DPM-Solver-3 overtakes DPM-Solver-1 at 14 NFE. Finally, trajectory accuracy and image quality disagree: at 10 NFE DPM-Solver-2 lands $2.3\times$ further from the converged image than DPM-Solver-1 yet scores $1.8\times$ better FID, because it produces a sharp but *different* picture. We also identify when DPM-Solver’s exact linear treatment helps (concentrated data) and when it hurts (spread-out data), and we report the predictions we got wrong.

1 The problem, in numerical terms

A diffusion model generates an image by starting from Gaussian noise and following a deterministic flow to the data manifold. That flow is the *probability-flow ODE*, and evaluating its right-hand side requires one forward pass of a large neural network. The network call is essentially the entire cost of sampling, so the governing question is how few evaluations one can afford before the trajectory leaves the manifold. The unit of cost is the NFE, the number of function evaluations.

The base paper’s contribution is a change of variables. The right-hand side splits into a linear part, which is exactly integrable in closed form, and a neural part, which is not. Classical explicit schemes approximate both at once and waste error budget on the half that was free. DPM-Solver treats the linear half exactly and spends its entire approximation on the hard half, reaching good samples in 10–20 NFE where prior samplers needed 100–250.

The paper proves an order theorem for this scheme and validates it with FID scores on generated images. FID is a distributional measure of image quality; it is not a measure of how accurately an ODE was integrated. Three quantities a numerical analyst would ask for are therefore measured only indirectly, if at all:

Order. Halving the step size h should reduce the error by 2^p for a p th-order method. The paper proves p but does not measure it against a known-correct trajectory.

Stability. The ODE stiffens as $t \rightarrow 0$ because $\sigma(t) \rightarrow 0$. §4.2 of the paper suggests, citing the exponential-integrator literature, that classical explicit methods “may suffer from unstable numerical issues for large step size”; it does not measure a stability limit.

Cost. A higher-order method is more accurate per step but calls the network more times per step. Whether it wins after dividing out is an empirical question.

This report supplies all three, and then puts the numerical verdict next to the image-quality one. An earlier draft compared measurements taken under different settings; a review caught it, and the controls in §§4–7 replace those comparisons. Where a conclusion changed as a result, the ledger in §8 says so.

2 Mathematical setting

We use the variance-preserving linear schedule of the base paper, with $\beta_0 = 0.1$, $\beta_1 = 20$, and t running from $T = 1$ down to a small t_{end} :

$$\log \alpha(t) = -\frac{\beta_1 - \beta_0}{4}t^2 - \frac{\beta_0}{2}t, \quad \sigma(t) = \sqrt{1 - \alpha(t)^2}, \quad (1)$$

$$\beta(t) = \beta_0 + t(\beta_1 - \beta_0), \quad f(t) = \frac{d \log \alpha}{dt} = -\frac{1}{2}\beta(t), \quad g^2(t) = \beta(t). \quad (2)$$

The half log-SNR $\lambda(t) = \log \alpha(t) - \log \sigma(t)$ is the natural time variable for this problem. It is *strictly decreasing* in t : $t = 1$ gives the most negative λ , and $t \rightarrow 0$ sends $\lambda \rightarrow +\infty$. Its inverse has a cancellation-free closed form,

$$t(\lambda) = \frac{2u}{\sqrt{\beta_0^2 + 2(\beta_1 - \beta_0)u + \beta_0}}, \quad u = \log(e^{-2\lambda} + 1). \quad (3)$$

Writing $\varepsilon(x, t)$ for the noise-prediction model — a network in general, an exact formula on our analytic tiers — the same trajectory admits three discretisable forms, and these define our three solver arms:

$$(A) \quad \frac{dx}{dt} = f(t)x + \frac{g^2(t)}{2\sigma(t)}\varepsilon(x, t), \quad \text{uniform grid in } t, \quad (4)$$

$$(B) \quad \frac{dx}{d\lambda} = \hat{\sigma}(\lambda)^2 x - \hat{\sigma}(\lambda)\varepsilon(x, t(\lambda)), \quad \text{uniform grid in } \lambda, \quad (5)$$

$$(C) \quad x_i = \frac{\alpha_i}{\alpha_{i-1}}x_{i-1} - \sigma_i(e^h - 1)\varepsilon(x_{i-1}, t_{i-1}), \quad h = \lambda_i - \lambda_{i-1}, \quad (6)$$

with $\hat{\alpha}(\lambda) = (1 + e^{-2\lambda})^{-1/2}$ and $\hat{\sigma}(\lambda) = (1 + e^{2\lambda})^{-1/2}$. Equations (4) and (5) are the same ODE in different coordinates; (6) is DPM-Solver-1, exact on the linear part. Comparing A with B isolates the reparameterisation alone; comparing B with C isolates the exact linear treatment alone. Figure 1 makes the distinction concrete.

We implement $e^h - 1$ as `expm1(h)` throughout. The naive form loses relative precision catastrophically for small h , which is exactly the regime a convergence study measures.

3 Experimental design

3.1 Three tiers

To measure error one needs the right answer, which for a real generated image does not exist. We therefore build deliberately simplified problems whose exact solution is available, and step up in realism (Figure 2).

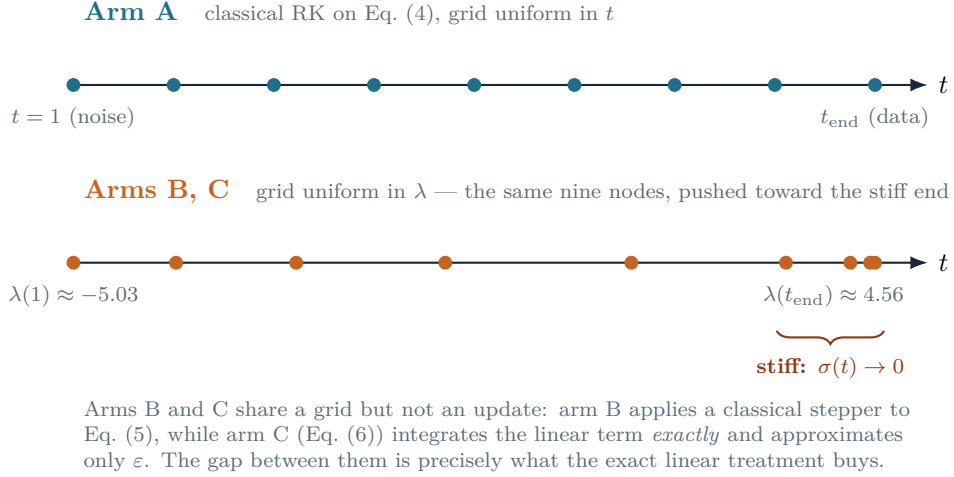


Figure 1: The three arms. All integrate one trajectory; they differ in the time variable that is discretised uniformly and in how much of the right-hand side is approximated. A uniform- λ grid automatically clusters nodes where the problem stiffens.

Tier 1 — anisotropic Gaussian. For data $\sim \mathcal{N}(0, \text{diag}(s))$ the ODE decouples into d independent scalar linear ODEs. Both the score and the entire trajectory are closed form:

$$\varepsilon^*(x, t)_j = \frac{\sigma(t) x_j}{v_j(t)}, \quad x_j(t) = x_j(T) \sqrt{\frac{v_j(t)}{v_j(T)}}, \quad v_j(t) = \alpha(t)^2 s_j + \sigma(t)^2. \quad (7)$$

The stiffness knob is the condition number $\kappa = s_{\max}/s_{\min}$, which we sweep over five decades. Because the linear case decouples, dimension changes nothing — we sweep κ , not d .

Tier 2 — Gaussian/point mixture. Replacing the single Gaussian by a weighted point cloud $\mu_1, \dots, \mu_K$ keeps the score closed form but makes it genuinely nonlinear in x : the posterior mean becomes a softmax over modes,

$$r_k \propto w_k \exp\left(-\frac{\|x - \alpha(t)\mu_k\|^2}{2\sigma(t)^2}\right), \quad \hat{x}_0 = \sum_k r_k \mu_k, \quad \varepsilon^*(x, t) = \frac{x - \alpha(t)\hat{x}_0}{\sigma(t)}. \quad (8)$$

We place $K = 8$ modes on a Swiss-roll curve. There is no algebraic trajectory, so the reference is SciPy’s DOP853 at $\text{rtol} = 10^{-13}$, $\text{atol} = 10^{-14}$, verified converged by recomputation at 10^{-11} (agreement 6.9×10^{-13} , far below every measured error). The responsibilities are computed with `logsumexp`; a naive exp-then-normalise overflows long before the ratios cease to be well defined.

Tier 3 — the real network. `google/ddpm-cifar10-32`, $d = 3072$, on a Kaggle T4 — the same DDPM checkpoint as the paper’s discrete-time CIFAR-10 experiments. The reference is a 201-NFE DPM-Solver-3 run of the *same* checkpoint from the *same* x_T , so what is measured remains discretisation error rather than model error. §7 adds what this metric cannot see: the samples themselves, and their FID.

3.2 Two protocols

A convergence study depends on *where* and *at what budget* it is run, and this report uses two settings deliberately (Table 1).

The asymptotic protocol avoids the region near the data, where the problem is hardest, and starts each sweep past the solver’s pre-asymptotic regime — the right conditions for checking that an implementation has the order the theorem promises. The practitioner protocol is Tier 3’s: a

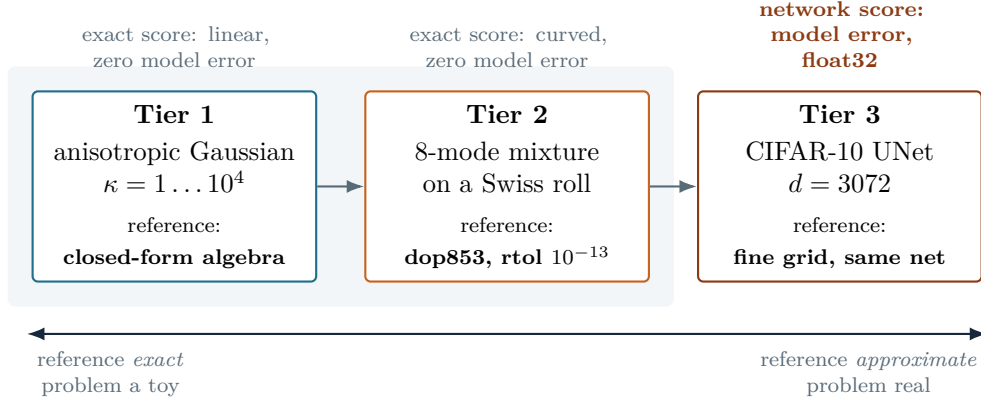


Figure 2: The tier ladder. Moving right trades exactness of the reference for realism of the problem. The shaded pair have analytic scores, so any shortfall from theory there is a property of the solver and the budget, not of a model.

Table 1: The two measurement protocols. Every cross-tier comparison in this report uses the practitioner protocol.

Protocol	Interval	Budget	Purpose
asymptotic	$t \in [0.2, 1]$	≥ 16 –64 NFE, per solver past its pre-asymptotic region	test the order theorem cleanly (Tiers 1–2, §4.1)
practitioner	$t \in [10^{-3}, 1]$ (the paper’s ϵ)	9–120 NFE	what a user runs; Tier 3’s only setting, and the one Tiers 1–2 are re-run at to be compared with it

network is expensive, and nobody samples it at 500 NFE. An earlier draft of this report compared Tier-1/2 slopes from the first protocol with Tier-3 slopes from the second, and attributed the difference to the network; §5.2 corrects that.

3.3 Conventions

Three decisions were fixed before any measurement and never revisited. **Float64 everywhere** on Tiers 1–2: a float32 error floor near 10^{-7} would destroy the order fits. **Every run seeded.** **One results schema** — every experiment appends a row (`tier`, `testbed`, `arm`, `solver`, `order`, `kappa`, `h`, `nfe`, `err_l2`, `diverged`, `seed`) to a CSV, so all tables concatenate without adapters.

Two accounting rules matter throughout. NFE is *not* step count: RK4 spends four network calls per step, DPM-Solver- k spends k , and AB2 spends one after startup. Every cross-arm comparison uses *measured* NFE, obtained from a call counter wrapped around the right-hand side rather than assumed from a formula. And order is defined in h , not NFE; since arms measure h in different units, h is used only *within* an arm.

3.4 Fitting an order

A real convergence curve has three regions: pre-asymptotic at large h where theory does not yet apply, a straight middle, and a round-off floor at small h . Fitting one line through all points averages across regimes and reports the wrong number. We instead search every contiguous log–log subwindow and keep the one that is both straight and wide, and we report the fit window and R^2 beside every slope. This procedure is not infallible — §5 documents a case where it must be told about the floor explicitly.

Table 2: Measured convergence order on the two exact-score testbeds, asymptotic protocol ($t_{\text{end}} = 0.2$). Slopes are log-log fits of $\|x_N - x_{\text{ref}}\|_2$ against h over the automatically selected asymptotic window. §5.2 repeats the measurement under the practitioner protocol.

Arm	Solver	Theory p	Tier 1	Tier 2	$ \Delta $ max
A	euler	1	1.012	1.012	0.012
A	midpoint	2	2.028	2.031	0.031
A	heun3	3	3.014	2.999	0.014
A	rk4	4	3.938	3.997	0.062
A	ab2	2	2.033	1.988	0.033
B	euler	1	1.023	1.023	0.023
B	midpoint	2	2.057	2.055	0.057
B	heun3	3	3.015	3.024	0.024
B	rk4	4	3.921	3.965	0.079
B	ab2	2	1.935	1.975	0.065
C	ddim	1	0.994	0.996	0.006
C	dpm1	1	0.994	0.996	0.006
C	dpm2	2	2.028	2.018	0.028
C	dpm3	3	3.094	3.084	0.094

3.5 Validation: Gate G1

Section 4.1 of the paper proves DDIM and DPM-Solver-1 are algebraically the same update. Two independent implementations must therefore agree to machine precision. Ours agree to 5.5×10^{-16} against a 10^{-14} bar, and DPM-Solver-1 measures at order 0.99. Everything downstream was gated on this. Separately, the Tier-1 closed-form trajectory satisfies the ODE to $\sim 10^{-11}$ by finite difference and matches an independent DOP853 integration to 10^{-12} ; our schedule agrees with the vendored `NoiseScheduleVP('linear')` to 10^{-11} , so all three arms provably share one schedule.

We use the authors’ own solver file, unmodified and pinned to a commit, driven by an analytic oracle. This tests the *published implementation* with zero model error rather than a reimplementa-tion of it.

4 Results on the analytic tiers

4.1 Order, in the asymptotic regime

Table 2 gives the measured slopes under the asymptotic protocol. Every solver in every arm attains its theoretical order: the worst deviation is 0.094 on Tier 1 and 0.084 on Tier 2, with mean R^2 of 0.99992 and 0.99996. Introducing genuine curvature (Tier 2) shifts no slope by more than 0.059.

This is the control experiment. Both tiers have analytic scores, so the smoothness hypothesis behind the paper’s order theorem (its Assumption B.1: the λ -derivatives of ε exist and are continuous) holds, and the theorem’s conclusion holds with it. It also confirms that the published solver code, driven by an exact oracle, is correct.

4.2 Stability: absent on the linear problem, real on the curved one

Section 4.2 of the paper suggests that explicit methods on this semi-linear ODE destabilise at large steps near $t \rightarrow 0$. We measured it directly, bisectioning on step count for the coarsest grid that stays bounded, for all five steppers in arms A and B, on both analytic tiers down to $t_{\text{end}} = 10^{-3}$.

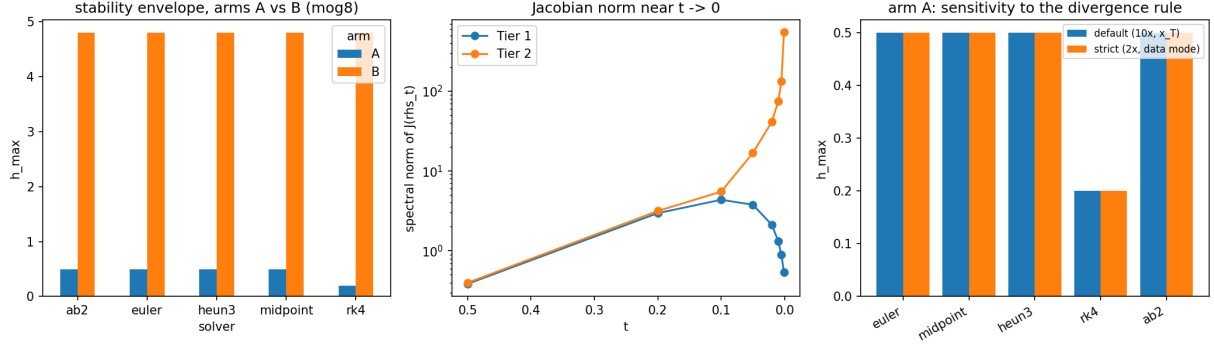


Figure 3: Stability on the curved Tier 2 (M12.2). Left: largest stable step for each stepper, arms A and B (a bar at 0.4995 in t or 4.79 in λ means “stable at the coarsest grid tried”). Middle: spectral norm of the arm-A Jacobian along the trajectory; the linear Tier 1 turns *down* near $t \rightarrow 0$, the curved Tier 2 climbs to 545. Right: the arm-A limits under the default and a stricter divergence rule — identical.

Tier 1: no instability at any κ . Across $\kappa \in \{1, 10, 10^2, 10^3, 10^4\}$, every stepper in both arms is already stable on the coarsest grid tried (2 steps: $h \approx 0.4995$ in t , 4.79 in λ), so the bisection never engages; stress-testing to $\kappa = 10^8$ and $t_{\text{end}} = 10^{-6}$ changes nothing. (The bisection itself is verified against a synthetic linear ODE with a closed-form instability threshold.) The reason is structural. On a uniform- t grid the node nearest the boundary sits at $t_{\text{end}} + h$, and the stability product $h \cdot J(t_{\text{end}} + h)$, with J the local Jacobian, never exceeds ≈ 0.9 for any h or κ — under explicit Euler’s threshold of 2 — because $\sigma(t)^2 \sim \beta_0 t$ near the boundary makes the growing stiffness and the grid’s shrinking distance to t_{end} very nearly cancel. The stiffness knob κ cannot defeat this: for $s_{\min} \ll \sigma^2$ the local rate is capped by $\sigma(t)^2$, not by $s_{\min}$.

Tier 2: a real limit, and only in t . The curved mixture behaves differently (Figure 3). Its Jacobian grows without that cancellation: between modes the posterior weights switch ever more sharply as $\sigma \rightarrow 0$, and at $t = 10^{-3}$ the spectral norm of $\partial(\text{rhs}_t)/\partial x$ is 545, against 0.54 on Tier 1 — three orders of magnitude. Arm-A RK4 now has a genuine stability limit, $h_{\max} = 0.1998$: stable from 5 steps, divergent at 2–4. A stricter divergence rule (twice the data scale instead of ten times $\|x_T\|$) gives the same limit. The other arm-A steppers stay stable at the coarsest grid, and the reason is where they sample: RK4 is the only one whose final stage evaluates the right-hand side at the *end* of the step — on the last step, at t_{end} itself, where the Jacobian peaks. In arm B every stepper is stable at every grid tried.

So the paper’s §4.2 claim is *narrowed*, not refuted: explicit steppers in t do destabilise near the data, but the cause is curvature of the data distribution, not the stiffness parameter the claim — and our original prediction — points to; and the λ -reparameterisation removes it.

4.3 When does the exact linear treatment help?

DPM-Solver’s premise is that integrating the linear term exactly and approximating only ε is always the better split. Under the practitioner protocol the two analytic tiers disagree about that:

- On Tier 1 ($\kappa = 10$), DPM-Solver-1 and -2 lose to the classical stepper of the same order, in either coordinate, at every budget from 10 to 120 NFE: at 20 NFE, DPM-Solver-2 errs by 0.133, RK2 in t by 0.034, RK2 in λ by 0.0089; at 10 NFE, DPM-Solver-1 errs by 0.425 against Euler-in- t ’s 0.092. DPM-Solver-3 loses to RK4 in λ throughout, and beats RK4 in t only above ~ 23 NFE.

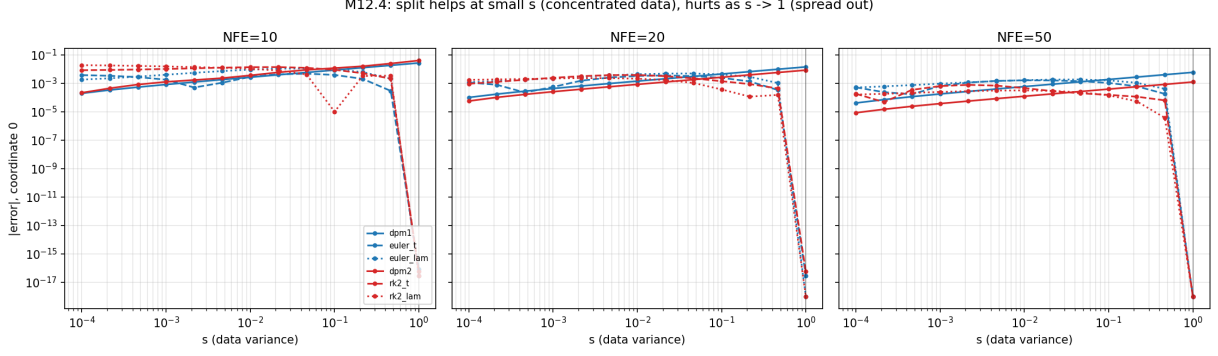


Figure 4: When the split helps (M12.4). Error on one Gaussian coordinate of data variance s , at 10, 20 and 50 NFE. Solid: DPM-Solver-1 (blue) and -2 (red); dashed/dotted: the same order in t and in λ . At small s DPM-Solver is ahead; at $s = 1$ the classical right-hand side cancels exactly and they drop to machine precision.

- On Tier 2 it wins where the paper says it should: at 12 NFE, DPM-Solver-2 errs by 0.084 against 0.23 (RK2 in λ) and 0.34 (RK2 in t), and DPM-Solver-3 by 0.049 against 0.30 (RK4 in λ) and 14.8 (RK4 in t : 3 steps, past its 5-step stability limit, §4.2).

The difference is how *concentrated* the data is, and it has a two-line explanation. DPM-Solver’s local error is driven by how much ε varies along the trajectory in λ ; a classical stepper’s is driven by the whole right-hand side. For data concentrated at a point ($s \rightarrow 0$), $\varepsilon = x/\sigma$ and the exact flow is $x \propto \sigma$, so ε is *constant* along the path and DPM-Solver-1 is exact; substituting $e^h = (\alpha_i \sigma_{i-1})/(\sigma_i \alpha_{i-1})$ into Eq. (6) gives $x_i = (\sigma_i/\sigma_{i-1}) x_{i-1}$ identically. For data as spread out as the noise ($s = 1$), $v \equiv 1$, the whole right-hand side of Eq. (4) vanishes and every classical stepper is exact, while $\varepsilon = \hat{\sigma}(\lambda) x$ still varies from 1 to 0 across the trajectory and DPM-Solver-1 is not. Figure 4 sweeps s between the two: at 20 NFE, for $s < 10^{-2}$ DPM-Solver-1 beats Euler-in- t at 83% of points and DPM-Solver-2 beats RK2-in- t at all of them; at $s = 1$ the classical steppers are exact to 10^{-16} while DPM-Solver-1 errs by 5.9×10^{-3} . Tier 1’s eigenvalues run up to $s = 1$; Tier 2 is a set of point masses. Natural images, whose pixel-space covariance spectrum decays rapidly, put most of their directions at the concentrated end, which is consistent with DPM-Solver’s success on them — though we have not measured that spectrum here.

5 Tier 3: the order theorem meets a real network

5.1 One schedule, read off the checkpoint

The CIFAR-10 checkpoint is a *discrete-time* DDPM: 1000 β values, with $\log \alpha$ linearly interpolated between tabulated times. That is the same VP-linear schedule discretised, and it is *not* interchangeable with the continuous form — we measure $\max |\Delta \lambda| = 4.74 \times 10^{-2}$ and a 5.0% discrepancy in f . Using continuous coefficients for arms A and B would place a systematic 5% error beneath every Tier-3 curve.

We therefore derive all three arms from the checkpoint’s own schedule. Arm B needs only $\hat{\sigma}(\lambda)$ — which follows from $\lambda = \log \alpha - \log \sigma$ and $\alpha^2 + \sigma^2 = 1$ alone, hence is exact for any variance-preserving schedule — and $t(\lambda)$. Arm A needs f and g^2 , and here one identity does all the work. Differentiating $\sigma^2 = 1 - \alpha^2$ gives $\sigma' = -\alpha^2 f/\sigma$, so

$$\lambda' = f - \frac{\sigma'}{\sigma} = f \left(1 + \frac{\alpha^2}{\sigma^2} \right) = \frac{f}{\sigma^2}, \quad \text{hence} \quad \boxed{f = \sigma^2 \lambda', \quad g^2 = -2\sigma^2 \lambda' = -2f.} \quad (9)$$

Taking *both* coefficients from a single central difference of λ makes $f/\lambda' \equiv \sigma^2$ hold to 1.1×10^{-16} rather than approximately — and that identity is exactly what makes (4) and (5) the same ODE.

Table 3: Measured order under the asymptotic protocol (Tiers 1–2 only) and the practitioner protocol (all three tiers). “Floor-excl.” refits Tier 3 after discarding sweep points within $3\times$ the measured reference floor (the guide’s “do not fit slopes through the flat part”). From `results/matched_protocol_slopes.csv`.

Arm	Solver	p	asymptotic		practitioner ($t_{\text{end}} = 10^{-3}$, 9–120 NFE)			
			Tier 1	Tier 2	Tier 1	Tier 2	Tier 3	floor-excl.
A	euler	1	1.012	1.012	0.871	1.269	0.842	0.842
A	midpoint	2	2.028	2.031	2.403	1.485	1.525	1.500
A	rk4	4	3.938	3.997	3.297	2.093	1.210	1.210
B	euler	1	1.023	1.023	1.073	1.191	0.746	0.746
B	midpoint	2	2.057	2.055	1.786	2.385	1.603	1.603
B	rk4	4	3.921	3.965	4.829	4.523	2.231	1.099 [†]
C	dpm1	1	0.994	0.996	0.958	1.325	0.813	0.813
C	dpm2	2	2.028	2.018	2.100	2.455	2.191	2.244
C	dpm3	3	3.094	3.084	3.492	4.214	2.473	2.995
worst $ \text{slope} - p $			0.094	0.084	0.83	1.91	2.79	

[†] B/rk4 loses two of eight points to the floor, leaving six that lie mostly in its pre-asymptotic region; neither Tier-3 number should be trusted for that row.

Arms A and B therefore agree by construction; we verify it end to end, integrating both at four resolutions and watching the gap fall from 2.5×10^{-4} to 3.9×10^{-7} .

5.2 Measured order at practitioner budgets

Table 3 puts the network’s slopes next to the analytic tiers’ — now under *both* protocols, so that like is compared with like.

Most of the shortfall is the budget, not the network. Read across a row. In the asymptotic regime the exact-score slopes sit on theory. Under the practitioner protocol the *same exact-score problems* scatter on both sides of theory by up to 1.9 — RK4 in t on Tier 2 measures 2.09, RK4 in λ 4.52. Nothing about the score changed; only the interval and the budget did. At 9–120 NFE, reaching down to $t_{\text{end}} = 10^{-3}$, no solver is in the asymptotic regime its order refers to, whether or not a network is involved. The exact-score Tier 2 already shows 68% of the worst gap the network shows (1.91 of 2.79).

What the network adds. Tier 3’s slopes are all *below* theory, and the high-order steppers lose most against the matched Tier-2 control: A/rk4 $2.09 \rightarrow 1.21$, B/rk4 $4.52 \rightarrow 2.23$, dpm3 $4.21 \rightarrow 2.47$. That downward shift is the network’s own contribution — smaller than an asymptotic-vs-practitioner comparison suggests, and not uniform. The one clear exception is DPM-Solver-3: discarding the single sweep point that reaches the reference floor restores 2.995, essentially its textbook order. The automatic window search would have missed that — it prefers a wider window and fits straight through a flat tail unless told where the floor is — and curves far from the floor (A/rk4 at $12\times$ it, B/euler at $42\times$) lose nothing to the exclusion.

On Assumption B.1. An earlier draft attributed the Tier-3 shortfall to the network violating the order theorem’s smoothness hypothesis. That is hard to defend: B.1 asks only that the λ -derivatives of ε *exist and be continuous*, and a UNet with SiLU activations, normalisation and sinusoidal time embeddings is a smooth function of its inputs. What a network can do is make those derivatives *large*, which pushes the start of the asymptotic regime to smaller step sizes than

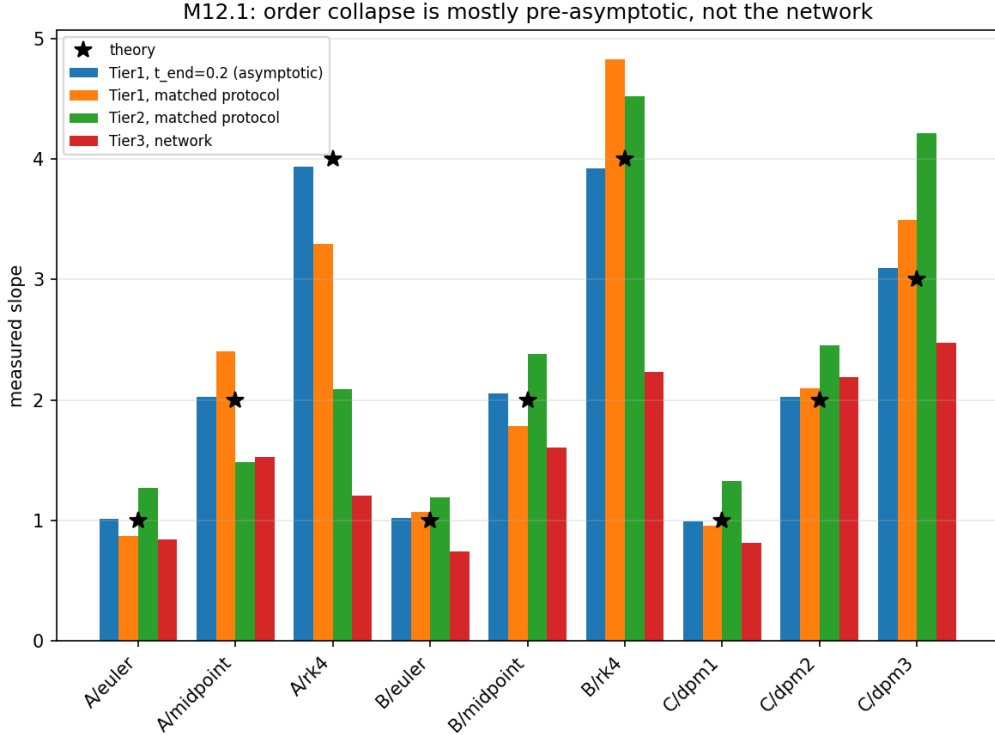


Figure 5: Measured slopes against theory (stars): Tier 1 under the asymptotic protocol, Tiers 1 and 2 under the practitioner protocol, and Tier 3. The large departures already appear with an exact score once the protocol matches Tier 3’s.

any practical budget reaches. The theorem is about $h \rightarrow 0$; at 10–120 NFE it is simply not the operative description, on any tier.

5.3 What the λ -reparameterisation buys

Arm A and arm B integrate the same ODE; they differ only in the time variable that is discretised uniformly. Table 4 gives the error of arm A over arm B at the largest budget both reach, so a value above 1 means the λ form is better.

Table 4: Error of arm A divided by error of arm B at the largest shared NFE (results/reparam_gain.csv). Above 1: the λ -uniform grid is more accurate.

Protocol	Tier	Euler	RK2	RK4
asymptotic ($t_{\text{end}} = 0.2$)	1	0.93	0.32	0.47
	2	0.92	0.39	1.77
practitioner ($t_{\text{end}} = 10^{-3}$)	1	0.69	0.85	48.4
	2	9.92	33.1	8551
	3	0.63	0.56	10.6

For RK4 the reparameterisation pays off on *every* tier once the interval reaches the data end — by $48\times$, $8551\times$ and $10.6\times$ — and not only on the network, as an earlier draft concluded from the asymptotic protocol, where it is worth $0.47\times$ and $1.77\times$. A λ -uniform grid places its nodes where the problem stiffens (Figure 1); a uniform- t grid leaves one long last step across the region where $\sigma \rightarrow 0$. For the low-order steppers the picture is mixed: on the curved Tier 2 the λ form helps by $10\text{--}33\times$, while on Tier 1 and on the network it is slightly worse ($0.56\text{--}0.85\times$). The reparameterisation matters most for exactly the steppers most exposed to the stiff end — the

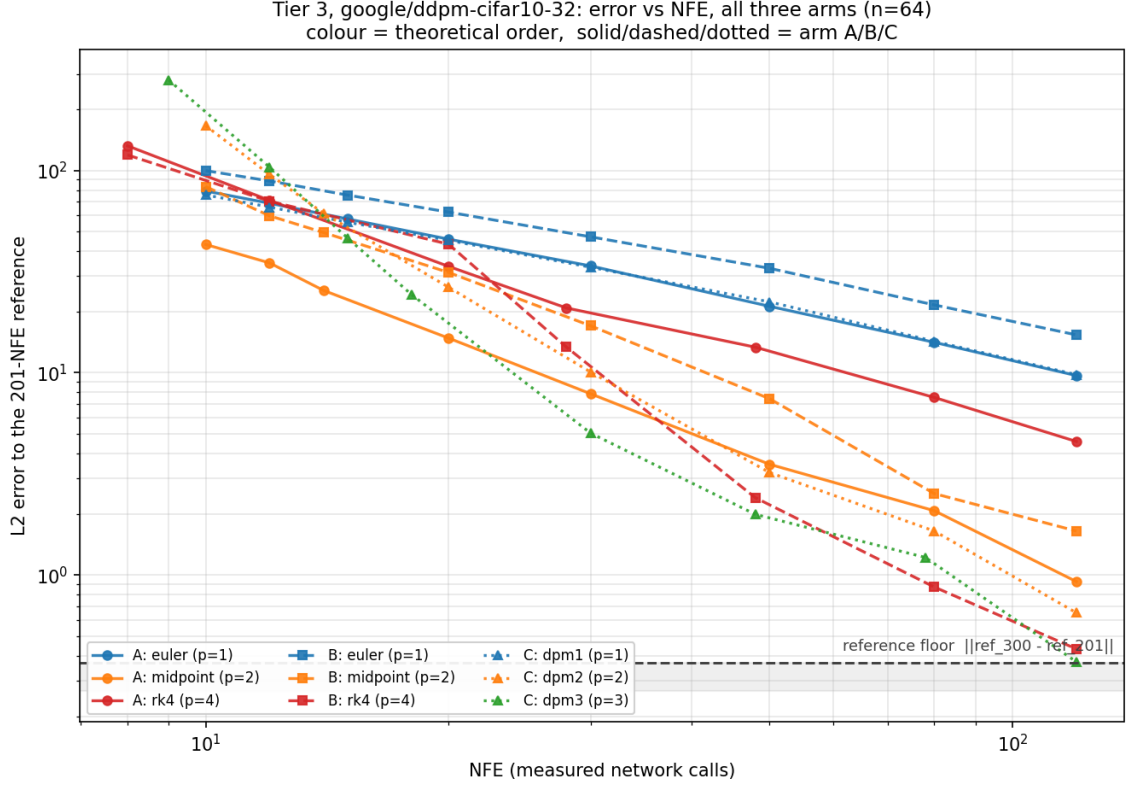


Figure 6: Tier 3: error against measured NFE for all nine solver/arm combinations. Colour encodes theoretical order, line style encodes arm. The dashed horizontal line is the reference’s own uncertainty $\|x_{300} - x_{201}\|$; nothing below it is a measurement of discretisation error. Note the separation between solid and dashed red (arm A versus arm B RK4).

high-order ones, whose stages sample it — which is the same mechanism behind RK4’s stability limit in §4.2.

5.4 The three-way error decomposition

Total Tier-3 error has three components, not one, and each is measurable without knowing the true score (Figure 7).

you control this	you choose this	you are stuck with this
discretisation $\ x_N - x_{\text{ref}}\ $	t_{end} truncation $\ x_{\text{ref}}(t_{\text{end}}) - x_{\text{ref}}(10^{-3})\ $	network + precision $\ x_{300} - x_{201}\ $
279.6 $\rightarrow$ 0.37 over 9–120 NFE	1.53/5.43/11.56 at $t_{\text{end}} = 2/5/10 \times 10^{-3}$	0.368 (irreducible)

The floor of 0.368 total is 4.6×10^{-2} per sample against $\|x\| \approx \sqrt{3072} \approx 55$, i.e. $\approx 8 \times 10^{-4}$ relative — matching the $\sim 10^{-3}$ float32 floor the base paper’s practitioners would predict, here measured rather than assumed.

The instructive comparison is between the second and first columns. Truncating at $t_{\text{end}} = 2 \times 10^{-3}$ instead of 10^{-3} moves the endpoint by 1.53 — more than DPM-Solver-3’s entire discretisation error at 78 NFE (1.22). Past that budget, refining the grid buys accuracy the truncation choice immediately throws away. This is the mechanism behind the base paper’s own Table 4, where FID *worsens* from 4.39 at 15 NFE to 5.52 at 20: the discretisation term has dropped below the truncation term, and the extra network calls are being spent on the wrong error.

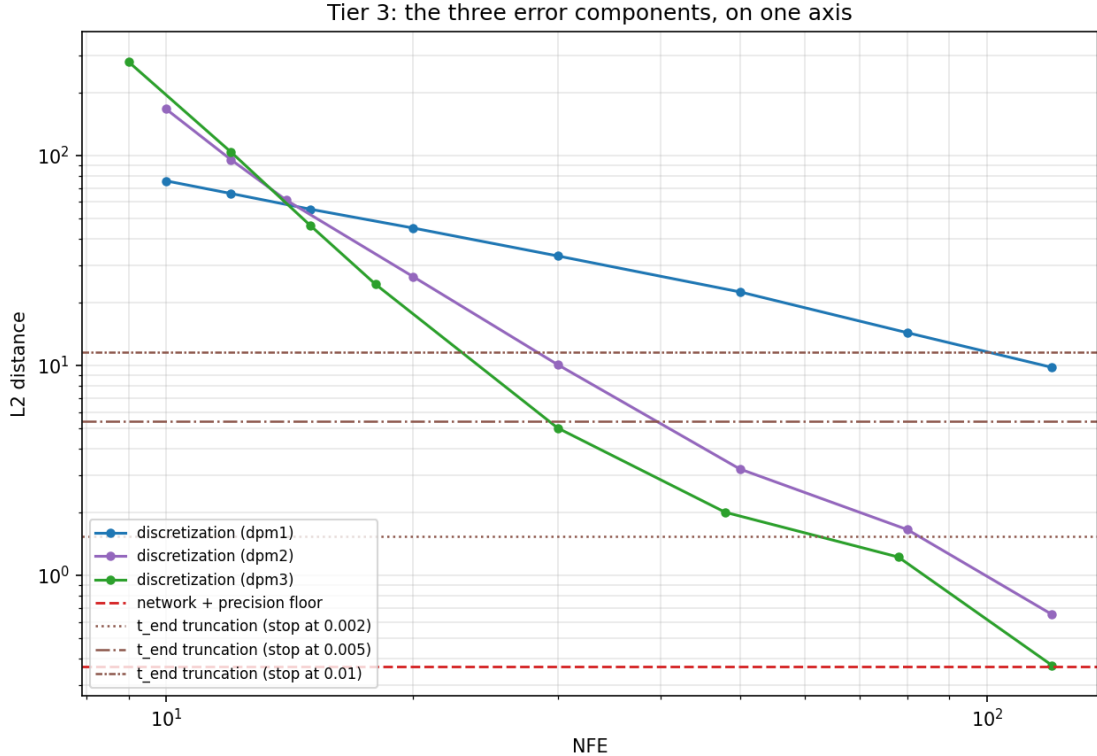


Figure 7: The three components on one axis. Where a discretisation curve crosses a horizontal line, that component ceases to dominate and further NFE stops buying end-to-end accuracy.

6 The crossover: when does order 3 beat order 1?

Table 6 of the base paper reports, without explanation, that on this very checkpoint (CIFAR-10 DDPM, discrete-time, $\epsilon = 10^{-3}$) DPM-Solver-3 is far *worse* than DPM-Solver-1 at 10 NFE — FID 24.37 (at an actual 9 NFE) against 16.69 — and far better by 12 NFE (8.20 against 13.63). This is the textbook pre-asymptotic picture: at large h the cubic Taylor correction is not yet valid and actively hurts. With exact answers and a continuously sweepable budget, the crossing can be located rather than guessed at.

It matters *what* is held fixed. Comparing the two orders at the same step size (M8) gives DPM-Solver-3 three times the budget; the crossing there sits at $\text{NFE}_3^* \approx 4.4\text{--}5.0$ on Tier 1, flat across five decades of κ , and 8.9 on Tier 2. The paper’s comparison — and the practical question — holds the *budget* fixed, so Table 5 compares at matched NFE (M12.3).

Table 5: NFE at which the DPM-Solver-3 and DPM-Solver-1 error curves cross, at matched NFE, practitioner protocol (`results/crossover_matched_nfe.csv`). Below the crossing order 1 is ahead; above it, order 3.

Testbed	Crossings (NFE)	Reading
Tier 1, Gaussian	5.1	one crossing
Tier 2, <code>mog8</code>	9.2, 14.2, 17.3	re-crosses; order 3 settled ahead from 17.3
Tier 3, CIFAR-10	14.1	one crossing — the paper’s own checkpoint

On the real network the crossing falls at 14 NFE: Table 6’s anomaly, reproduced on the paper’s checkpoint with a trajectory-error metric rather than FID. The first crossings rise with the difficulty of the problem — 5.1 for the linear Gaussian, 9.2 for the curved mixture, 14.1 for the network — which is what the pre-asymptotic reading predicts: the rougher ϵ is along the

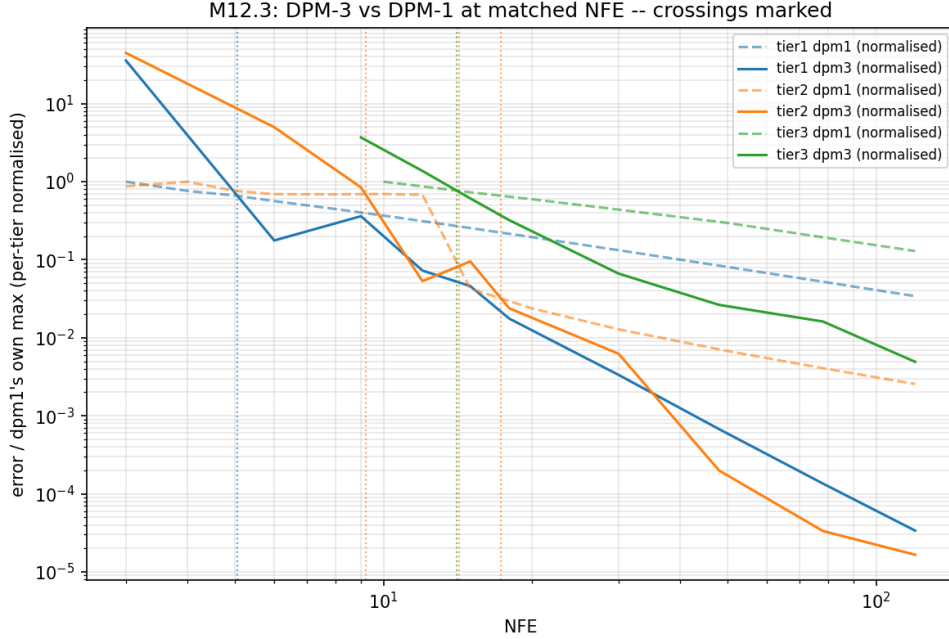


Figure 8: DPM-Solver-3 (solid) against DPM-Solver-1 (dashed) at matched NFE, each tier normalised by its own DPM-Solver-1 maximum; dotted verticals mark the crossings.

path, the larger its higher λ -derivatives, and the smaller the step must be before the extra Taylor terms help rather than hurt. Our own FIDs (§7) tell the same story: DPM-Solver-3 scores 143 at 9 NFE against DPM-Solver-1’s 44.9 at 10, and 18.7 at 18 against 27.0 at 20. Figure 8 shows the curves.

7 What the samples look like, and how FID ranks them

Everything so far measures how far a sampler’s endpoint lands from the converged ODE solution. The paper instead measures whether the images look right, with FID. M11 did both on the same checkpoint: it rendered samples from every sampler at every budget (Figure 9), and scored six samplers with FID on 5,000 samples each at ~ 10 and ~ 20 NFE, next to the L2 distance of the same samplers on a 64-image batch (Table 6). Two samplers the earlier milestones had not run are included because the paper itself uses them: *DPM-Solver-fast*, which mixes orders 1–3 to spend the budget exactly and is the paper’s sampler at ≤ 20 NFE, and DDIM on its customary quadratic time grid.

Our FID ranking matches the paper’s. At ~ 10 NFE the paper’s three headline comparisons all hold on our run: DPM-Solver-2 beats DPM-Solver-1 (24.9 vs 44.9), DPM-Solver-3 loses to DPM-Solver-1 (143 vs 44.9), and DPM-Solver-fast beats DDIM (31.3 vs 39.5). Only the top pair swaps order, DPM-Solver-2 edging DPM-Solver-fast where the paper has the reverse (6.42 vs 7.90 there, a small gap). This is the check that our Tier-3 pipeline reproduces the paper’s setting, and it passes.

Trajectory accuracy and image quality disagree. Among the five samplers that produce an image at all at 10 NFE, ranking by FID is the *exact reverse* of ranking by distance to the converged image: L2 orders them RK2-in- t , DPM-Solver-1, DDIM, DPM-Solver-fast, DPM-Solver-2; FID orders them DPM-Solver-2, DPM-Solver-fast, DDIM, DPM-Solver-1, RK2-in- t . DPM-Solver-2 lands $2.3\times$ further from the converged image than DPM-Solver-1 (180.6 vs 78.4) yet scores

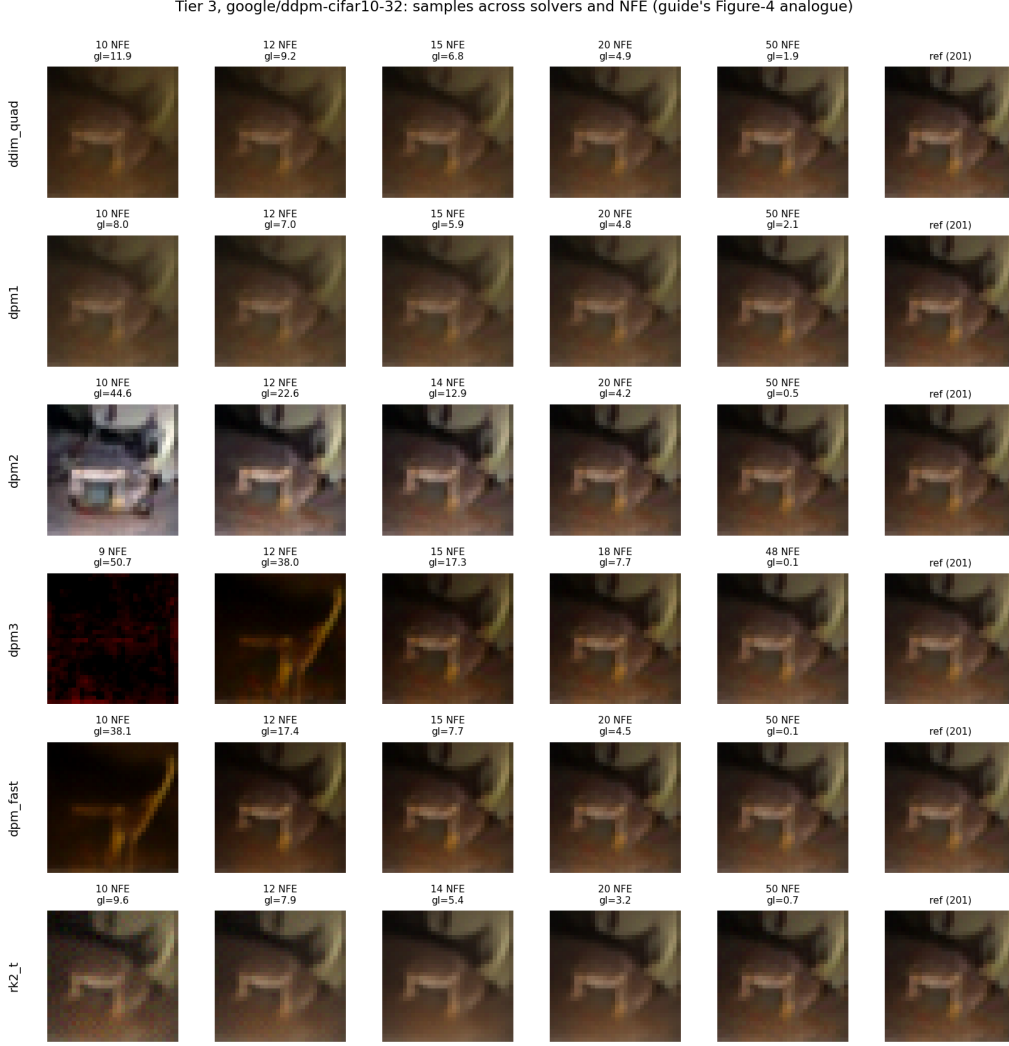


Figure 9: One starting noise, six samplers, five budgets, and the 201-NFE reference (right). “gl” is the mean absolute difference from the reference in 8-bit gray levels. At 10 NFE: DDIM and DPM-Solver-1 give the right picture, blurred; RK2 in t the right picture with residual noise; DPM-Solver-2 a sharp but *different* picture; DPM-Solver-fast an incomplete one; DPM-Solver-3 static. By 20 NFE every sampler has converged to the reference visually.

1.8 $\times$ better FID; M9’s independent 64-image batch agrees (166.4 vs 75.7). Figure 9 shows what happens, and suggests why: a few large steps at high noise decide the picture’s layout early and commit to it. DPM-Solver-2 then denoises crisply — a good sample, but not *this* trajectory’s sample — while RK2 in t stays on the trajectory and leaves it noisy. FID rewards the first, L2 the second. By ~ 20 NFE the two metrics largely agree again (rank correlation +0.60, against -1.00 at 10 NFE), and the best samplers are within FID-5k noise of the converged run (two score marginally *below* it).

This is the project’s thesis in one table. “DPM-Solver produces good images in 10 NFE” and “DPM-Solver integrates the ODE accurately in 10 NFE” are different claims; the paper’s evidence is for the first. For the second, no sampler we ran comes within a few gray levels of the converged image at 10 NFE: the closest, RK2 in t , is still 11 gray levels RMS away. Neither metric alone answers which sampler is best: one asks for a plausible image, the other for the image this noise was meant to become.

Table 6: Sample quality and trajectory accuracy on the same checkpoint. FID on 5,000 samples (clean-fid, CIFAR-10 training statistics); the paper’s FID for the same sampler (Table 6, DDPM discrete, $\epsilon = 10^{-3}$; DDIM from Table 4, 50,000 samples); L2 distance to the converged image over 64 images, and its per-image RMS in gray levels (median). Sorted by our FID within each budget. From `results/tier3_fid_vs_l2.csv`.

Budget	Sampler (NFE)	FID-5k	paper FID	L2 (64 img)	RMS gray
~10	DPM-Solver-2 (10)	24.9	7.90	180.6	49.3
	DPM-Solver-fast (10)	31.3	6.42	111.7	30.2
	DDIM, quadratic (10)	39.5	13.58	91.9	25.8
	DPM-Solver-1 (10)	44.9	16.69	78.4	21.9
	RK2 in t (10)	52.9	—	44.4	11.3
	DPM-Solver-3 (9)	143.3	24.37	287.0	82.5
~20	DPM-Solver-2 (20)	15.6	5.24	27.3	7.1
	RK2 in t (20)	15.7	—	15.6	3.7
	DPM-Solver-fast (20)	18.0	5.52	18.7	4.1
	DPM-Solver-3 (18)	18.7	5.43	26.4	6.4
	DDIM, quadratic (20)	25.1	6.94	49.8	13.9
	DPM-Solver-1 (20)	27.0	8.90	46.3	12.2
converged	DPM-Solver-3 (99)	16.5	~5.25	—	—

Caveats: absolute FID, and the sample-size floor. Our absolute FIDs are about three times the paper’s — even the converged run scores 16.5 against the paper’s ~5.25 — so only rankings are compared. How much of that is the sample size alone is measurable (notebook 11b, Figure 10): real, held-out CIFAR-10 test images, scored with the identical FID call, follow $\text{FID} \approx 3.0 \times 10^4/N$ almost exactly from 1,000 to 10,000 images. At 10,000 they give 3.15, clean-fid’s documented test-versus-train value, which also validates the FID pipeline itself. At our 5,000 images the floor — the best FID any sampler could score in this setup — is 5.9; at the paper’s 50,000 the same term is ≈ 0.6 . Sample size therefore explains 5.2 of the 11.2-point gap to the paper, a little under half. The other half is a real difference: with the sample-size term removed, our converged samples score ≈ 10.6 against the paper’s ≈ 4.6 . Plausible causes are clean-fid’s port of the TensorFlow Inception network and the discrete-to-continuous time conversion (the paper’s “Type-2 discrete” against the vendored code’s interpolated schedule); we did not isolate them. Because the floor shifts every sampler by the same amount, it changes no ranking in Table 6.

Two smaller points. The FID samples were seeded from Python’s per-process string hash, so they are not bit-reproducible and differ between samplers; differences of a few tenths are noise. The L2 columns use M11’s own 64-image batch (seed 1); M9’s seed-0 batch reproduces them to within 3–9%, with the same rankings.

8 The ledger

Predictions were written down before the measurements that test them: six in the project guide, before any sweep ran, and eight in the review that led to M11 and M12. Every verdict is computed from the results CSVs by `notebooks/10_report.ipynb` (`results/predictions_ledger.csv`), not asserted by hand. Reporting the misses is more useful than reporting only the hits.

Departures from the proposal. The efficiency frontier is reported in NFE rather than wall-clock (the paper’s Table 7 shows runtime is linear in NFE and near-identical across solvers, so wall-clock adds nothing). Tier 1 sweeps κ at fixed d rather than $d = 2 \dots 100$, because the linear case decouples and dimension changes nothing. Tier 2 uses a point mixture rather than a continuous Swiss roll, which has no closed-form score. The reference tolerance is 10^{-13} rather

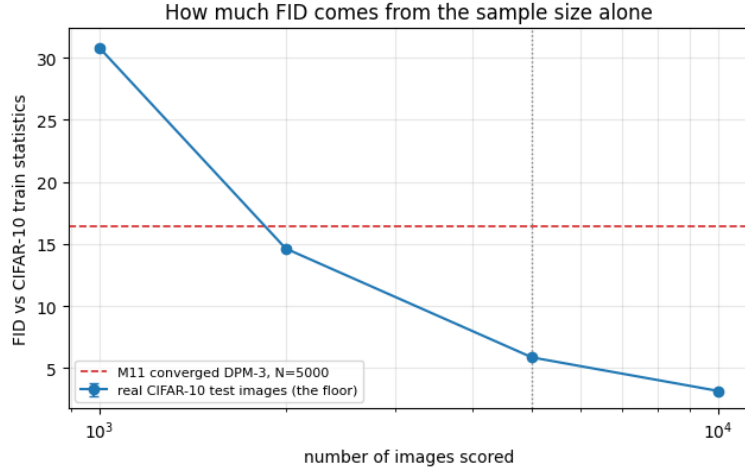


Figure 10: The FID floor (notebook 11b). Real, held-out CIFAR-10 test images scored with M11’s FID call, three random draws per size (the error bars are narrower than the markers). The dashed line is M11’s converged DPM-Solver-3 at 5,000 samples; the dotted vertical marks M11’s sample size.

than 10^{-14} , which SciPy refuses. Adaptive RK45 was **dropped**: step size is not a free variable for an adaptive method, so neither “order versus h ” nor “maximum stable h ” is defined on it; AB2 restores the multistep row instead. FID was cut at the week-2 gate and **restored in a limited role**: FID-5k on six samplers, as an anchor to the paper’s Table 6 and a counterpoint to L2 — never as the grading metric. Added beyond the proposal: arm B, Gate G1, the crossover study at both matched step and matched NFE, the three-way Tier-3 error decomposition, the matched-protocol controls, the stability study on a curved problem, the data-concentration sweep, and the first sample images.

Table 7: The guide’s six predictions.

Prediction	Verdict	Evidence
Tiers 1–2 slopes match theory within ± 0.15	confirmed	worst $ \Delta = 0.094$ over 28 fits, asymptotic protocol; but up to 1.91 under the practitioner protocol (R1)
Arm A degrades as κ grows; arms B/C do not	narrowed	no dependence on κ at all on the linear tier; on the curved tier arm-A RK4 has a real limit and arm B none — curvature, not κ (R2)
Euler- λ and DPM-Solver-1 are both first order but not identical	confirmed	slopes 1.02 vs 0.99; errors differ by $1.4\times$ at matched NFE
RK4 wins per step, loses per NFE, to DPM-Solver-3	narrowed	holds on Tier 3 for NFE 16–120 only; on Tiers 1–2 (asymptotic) RK4 wins both
Tier-3 slopes come out lower than theory	confirmed	8 of 9 fits > 0.15 below theory; most of the gap is the budget, not the network (R1)
Arm B at low NFE may be worse than arm A	confirmed	at the coarsest shared budget on all three tiers

Table 8: The review’s eight predictions, tested by M11, M12 and notebook 11b.

	Prediction	Verdict	Evidence
R1	Under Tier 3’s settings the exact-score tiers also miss theory: the shortfall is mostly the budget	confirmed	worst gap 1.91 (Tier 2) vs 2.79 (Tier 3) — 68% with no network (§5.2)
R2	On the curved tier, RK in t destabilises; the λ form does not	confirmed	arm-A RK4 $h_{\max} = 0.1998$; arm B: 0 of 5 limited; Jacobian $\sim 1000\times$ Tier 1’s (§4.2)
R3	At $t_{\text{end}} = 10^{-3}$ the λ form pays off for RK4 on every tier	confirmed	$48\times$, $8551\times$, $10.6\times$; mixed for Euler/RK2 (§5.3)
R4	At matched NFE, order 3 overtakes order 1 near the paper’s 10–12 NFE on its checkpoint	confirmed	crossing at 14.1 NFE; 5.1 and 9.2 on the analytic tiers (§6)
R5	The exact linear split helps on concentrated data, hurts on spread-out data	confirmed	$s < 10^{-2}$: DPM ahead at 83–100% of points; $s = 1$: classical steppers exact (§4.3)
R6	Our FID ranking at 10 NFE matches Table 6	confirmed	all three headline comparisons hold; top pair swapped (§7)
R7	L2 and image quality disagree at low NFE	confirmed	DPM-2 vs DPM-1: $2.3\times$ worse L2, $1.8\times$ better FID; rankings exactly reversed (§7)
R8	Most of our absolute-FID gap to the paper is the 5k sample size	narrowed	about half: 5.2 of 11.2 points (real images follow $\text{FID} \approx 3.0 \times 10^4/N$); the rest is a real pipeline difference (§7)

9 Reproducibility

The engine is plain Python under `src/`, unit-tested (129 tests), with no GPU or plotting code. Notebooks import it, run the sweeps and write `results/` and `figures/`. `run_all.sh` reproduces every laptop-side figure and CSV from a clean clone (notebooks 01–08, 10 and 12); it skips notebooks 09, 11 and 11b, which need a CUDA GPU and run on a Kaggle T4, and whose outputs are committed so that the report notebook reads them without a GPU.

One implementation note worth recording: the same integrator and the same right-hand side definition march all three tiers. `integrate` is backend-agnostic (float64 NumPy on Tiers 1–2, float32 CUDA tensors on Tier 3) and the probability-flow ODE is written once, parameterised by a schedule object. The claim that arms A and B mean the same thing across tiers is therefore structural rather than a matter of careful duplication.

10 Conclusion

Judged as a numerical method, DPM-Solver holds up where its theory applies, and the interesting findings are about where it does not.

On testbeds with analytic scores, in the asymptotic regime, every solver we tested reaches its textbook order to within 0.1, and the published code implements the method correctly. But the order theorem is a statement about $h \rightarrow 0$, and at the budgets diffusion models are sampled at — 10–120 NFE, integrated down to $t_{\text{end}} = 10^{-3}$ — no solver is in that regime, with or without a network: exact scores already miss theory by up to 1.9, most of the 2.8 seen on the network. At those budgets three other things decide which sampler wins.

Curvature. It is what destabilises explicit steppers near the data — not the stiffness knob the paper’s §4.2 and our own prediction pointed to — and it moves the crossover at which order 3 starts to beat order 1 later, from 5 NFE on a linear problem to 14 on the real network, where it reproduces the paper’s unexplained Table-6 anomaly. The λ -reparameterisation removes the instability and, for RK4, cuts error by one to four orders of magnitude on every tier.

How concentrated the data is. DPM-Solver’s exact linear treatment is exact for a point mass and a handicap for data as spread out as the noise, where the classical right-hand side cancels identically. Its success on images is consistent with image data being concentrated; it is not unconditional.

What “good” means. At 10 NFE ranking samplers by closeness to the converged image and ranking them by FID give exactly opposite orders among the ones that produce an image at all. DPM-Solver’s advantage at low NFE is real, and our FIDs rank it as the paper does; but it is an advantage in producing a *plausible* image, not in integrating the ODE. A practitioner who needs the latter — reproducible trajectories, interpolation in latent space, inversion — should budget on the order of 30–50 NFE: by 20 NFE the two rankings already largely agree (rank correlation +0.60, against -1.00 at 10), and DPM-Solver-3 comes within one gray level of the converged image at about 36.

Base paper: C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, J. Zhu. *DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps*. NeurIPS 2022, arXiv:2206.00927. Solver code MIT-licensed, vendored unmodified at a pinned commit.